The Distribution and Consequences of Natural Disasters: Evidence from U.S. Wildfire Victims

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Abstract

The economic burden of natural disasters is shaped by who is exposed, who experiences losses, how those losses alter individual economic trajectories, and how affected communities evolve in the aftermath. We link property-level damage data from over 100 U.S. wildfires to administrative tax and Census data to understand individual level relationships between disaster damage and economic outcomes. Residents of burned areas are wealthier than the U.S. population on average, but within these wildfire perimeters the likelihood that a home burns is negatively correlated with pre-fire income. Home destruction imposes costs beyond rebuilding the property: occupants of destroyed homes experience lower income for three years post-fire, producing cumulative income losses equal to 26 percent of pre-fire annual income. We do not detect income or employment effects for residents without direct property damage. Community recovery is slow and accompanied by changes in composition. Four years after a fire, half of destroyed addresses remain unoccupied and reoccupied addresses house residents with substantially higher incomes than before.

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1 Introduction

Global economic losses from natural catastrophes totaled $320 billion in 2024, continuing a trend of increasing costs.[1] Such aggregate statistics establish the scale of the challenge, but effective responses require more detailed evidence on who bears these losses, the extent to which financial disruptions extend beyond rebuilding costs, and how affected neighborhoods evolve during recovery. We study these dynamics in the context of catastrophic wildfire events in the United States. These events expose neighboring households to a common hazard, but damage outcomes are typically bimodal: homes either survive or are destroyed outright (Cohen 2000; Baylis and Boomhower 2026). For households whose homes are destroyed, recovery implies ground-up reconstruction rather than repair, with potential consequences for residents' economic trajectories and the composition of affected neighborhoods.

We combine address-level damage data for more than 100 U.S. wildfires between 2003 and 2022 with individual tax returns and Census microdata to reveal detailed demographic information about wildfire-affected households and the effects of wildfire events on earnings, employment, and migration in subsequent years. The highly granular damage data allow us to separately examine, and follow over time, the occupants of destroyed homes, homes that were exposed to the fire but were not destroyed, and unexposed homes located near the area of the fire.

Our first contribution is to describe the distribution of hazard and vulnerability. We document disparities in wildfire *hazard* by comparing the income, race, ethnicity, and age of the occupants of wildfire-affected (destroyed and surviving) homes to the US population and to nearby unaffected populations. Prior work relates predicted wildfire probabilities from wildfire models to tract-level measures of income and race (Davies et al. 2018; Wibbenmeyer and Robertson 2022). Our address-level data instead observe realized exposure and destruction, allowing us to characterize separately who is exposed and, conditional on exposure, whose home survives. We find that, on average, individuals affected by wildfires have higher income and are more likely to be white than the US population, reflecting the geography of wildfire risk. While the average income is relatively high, the population of wildfire-exposed individuals has a wide range of incomes, with 25% of occupants of destroyed homes having incomes below $42,000 per year ($2022).

We then examine disparities in structure *vulnerability* conditional on exposure to a wildfire. These analyses compare demographics for the occupants of destroyed vs. surviving homes. We find a clear income gap, consistent with earlier evidence of differences in structure sur-

1. Munich RE, "Natural Disasters in 2024", January 2025.

vival according to property characteristics (Baylis and Boomhower 2026; Plantinga, Walsh, and Wibbenmeyer 2022), and with disparities in ex ante wildfire risk mitigation (Anderson, Plantinga, and Wibbenmeyer 2023; Sølvsten, Whitlock, and Liao 2024; Reining et al. 2025). This evidence on demographic patterns in hazard and vulnerability provides valuable insights for the design and evaluation of risk mitigation initiatives and post-disaster assistance programs.

Our second contribution is to estimate the effects of U.S. wildfires on income, employment, and migration for affected individuals in the years following each event. Our empirical strategy considers several groups of individuals. The first group are people whose homes were destroyed in the wildfire, whom we refer to as "wildfire victims". The second group are people whose homes were inside the wildfire perimeter but were not destroyed (henceforth, "surviving homes"). We compare both groups to various comparison sets of households located away from the wildfire-affected area.

Relative to households outside of the burned area, average income for wildfire victims falls by 10% in the two years following the wildfire and remains below the level of the comparison group until the fourth post-fire year. In total, a typical wildfire victim foregoes the equivalent of 26% of their pre-fire single-year adjusted gross income across the years following the event. These income reductions represent a substantial increase in the financial impact for a typical wildfire victim on top of the uninsured portion of property damages. Among occupants of destroyed homes, there is a modest effect on long-distance migration. Losing one's home increases the likelihood of moving to a different county by about 10 percentage points. When these moves occur, they appear to be persistent for at least four years following the event. These estimates extend existing evidence that wildfires induce migration (Winkler and Rouleau 2021; McConnell et al. 2024; Hennighausen and James 2024; Wang 2025) by further isolating the response to direct home destruction for a sample of fires across several states and documenting its persistence.

Finally, we contribute new evidence on how neighborhoods change following a destructive wildfire. This analysis shifts focus from affected individuals to affected addresses, considering how the occupant mix evolves for destroyed, surviving, and adjacent homes. We find that rebuilding proceeds slowly and unequally. On average, 50% of destroyed homes remain unoccupied four years after a wildfire. Incomes among occupants of rebuilt homes are noticeably higher than the pre-fire average, and the share of occupants over age 65 is notably lower. These results are consistent with changes in the housing stock in burned neighborhoods reported by Issler et al. (2024).

A key advantage of our study is the ability to precisely observe the level of property damage

experienced by each individual resident. Previous studies of the financial consequences of floods, storms, and tornadoes proxy for individual damages with average damages by zip code or Census geography.[2] These geographic averages cannot distinguish households that suffer direct property loss from nearby households whose homes survive, making it difficult to isolate the resulting economic consequences. Both flood damages (Hovekamp and Wagner 2023) and wildfire damages (Baylis and Boomhower 2026) have been shown to vary substantially within small geographic areas, partly reflecting differing levels of defensive investment at the property level. The literature on flood and hurricane damages is further challenged by the fact that researchers often observe *predicted* flooding from a hydrological event model, as opposed to realized flooding on the ground.

A second advantage is the ability to observe annual income and employment from individual tax returns over two decades. The panel follows individuals across locations, allowing us to observe their outcomes after they leave the wildfire area. With the exceptions of Deryugina, Kawano, and Levitt (2018) and Groen, Kutzbach, and Polivka (2020), both of which study a single mega-disaster (Hurricane Katrina), we are not aware of any other papers that use income data from tax returns to understand the financial consequences of natural disasters in the United States.[3] Instead, the existing literature focuses on credit outcomes like delinquencies and debt balances (Gallagher and Hartley 2017; Billings, Gallagher, and Ricketts 2022; Gallagher, Hartley, and Rohlin 2023; Biswas, Hossain, and Zink 2023; Collier et al. 2024; Kahn, Ouazad, and Yönder 2024; Issler et al. 2024; An, Gabriel, and Tzur-Ilan 2024) or on spatially aggregated income and employment data (Roth Tran and Wilson 2025).

The income reductions that we measure for wildfire victims are larger and more persistent than those caused by Hurricane Katrina. One potential explanation for this difference may be that wildfire-damaged properties are usually a total loss and require ground-up reconstruction and complete replacement of household belongings, often accompanied by a period of extended temporary accommodation. In contrast, floods and wind storms are more likely to cause small or moderate amounts of damage. The result that income effects differ between hurricanes and other disasters is consistent with findings in Roth Tran and Wilson (2025), who study the effects of presidentially-declared disasters on county-level average income.

Taken together, our findings show that the economic incidence of wildfire cannot be inferred from exposure alone. It depends on which properties are destroyed, how those

2. For examples, see Gallagher and Hartley (2017), Deryugina, Kawano, and Levitt (2018), Billings, Gallagher, and Ricketts (2022), and Gallagher, Hartley, and Rohlin (2023).

3. Kabore, Rivers, and Armstrong (2025) study the 2011 Slave Lake wildfire in Canada using administrative data that do not distinguish between destroyed and surviving properties.

losses alter residents' subsequent economic trajectories, and how the composition of affected communities changes during recovery. Accounting for disaster costs therefore requires following both people and places beyond the event itself.

The paper proceeds as follows. Section 2 describes the construction of the dataset. Section 3 summarizes results related to wildfire hazard and vulnerability. Section 4 reports estimates of the causal effect of wildfire property loss on earnings, employment, and migration. Section 5 reports results for neighborhood-level changes and post-disaster gentrification. Section 6 discusses and contextualizes our findings, and Section 7 concludes.

2 Data

We construct a new individual-level panel dataset on wildfire property damages, occupant demographics, and post-incident economic outcomes. Sections 2.1 and 2.2 describe the input datasets and Section 2.3 describes construction of the merged dataset.

2.1 Wildfire Property Damage

Our data on address-level wildfire damages build on the dataset created in Baylis and Boomhower (2026) and are described in detail in that paper. The dataset includes structure-level damage outcomes for homes located inside approximately 150 wildfire burned areas in five western U.S. states. The data represent more than 70% of all U.S. wildfire-destroyed homes between 2003–2022 and more than 85% of wildfire-destroyed homes in the included states (California, Arizona, Colorado, Oregon, and Washington). The data were assembled by requesting and harmonizing individual post-incident damage censuses conducted by county and state emergency response officials. In Section 6, we also present auxiliary analyses of properties destroyed during the January, 2025 Los Angeles wildfires, which are outside the time period of the Baylis and Boomhower (2026) data. Damage data for the 2025 wildfires come from the DINS dataset provided by CAL FIRE.[4]

2.2 Tax and Census Data

Individual-level demographic characteristics and economic outcomes come from the Environmental Impacts Frame (EIF), a dataset assembled for environmental social science research by researchers affiliated with the United States Census Bureau. The EIF draws on restricted-access microdata from the universe of tax returns filed with the Internal Revenue

4. The DINS data are provided through a live data feed here: https://data.ca.gov/dataset/cal-fire-damage-inspection-dins-data. The data used for the 2025 Los Angeles wildfires analysis were obtained from this feed on February 21, 2025.

Service and multiple other administrative records sources (including data from Medicare enrollment and US Postal Service address changes). The EIF identifies a household's home location in a given year, covering the period 1999-2024 currently. Location in the EIF is determined by the location in the tax filing unless a more recent address is available from another administrative records source, such as Medicare enrollment data. The construction and contents of the EIF are described in more detail in Voorheis et al. (2026). Income and employment information is drawn from the Opportunity Databank, which harmonizes a similar set of IRS 1040 income and W-2 wage and employment records used in the EIF data from 1999-2022 (Chetty et al. 2026).

The main analysis focuses on several key variables from the EIF. For each individual in the data, we observe race and ethnicity. Annually, we observe their age, their adjusted gross income (AGI), wages, and retirement account withdrawals. AGI is our primary income measure, and includes labor and self-employment income, taxable transfer payments, and retirement withdrawals. We also include indicators for whether the individual reported any wages that year, if they changed jobs, and whether they reported owning their primary residence. We use the EIF location to determine if each individual moved from their original location, out of county, or out of state.

2.3 Construction of the Merged Dataset

We merge damaged homes in the wildfire data to addresses in the EIF data using a combination of geographic coordinates and address data. This combination allows us to link the data to the Master Address File Auxiliary Reference File, an internal dataset within Census that identifies individual-address pairs for a given year. We then link the individual identifiers, which are known as Protected Identification Keys or PIKs, to obtain the income and demographic data. This process is described in more detail in Voorheis et al. (2026).

We apply the following sample exclusions. First, the merged dataset is limited to occupants of single family homes. This is the group for which we are able to merge the damage data to the Master Address File Auxiliary Reference File with the greatest confidence. Single family homes comprise 88% of destroyed residences in the wildfire damage assessment data. Second, for the subset of analyses related to earnings, we restrict this sample to individuals between 25 and 65 years of age.

The main analysis dataset includes five categories of individuals for each incident, classified by the location of their homes at the time of a given wildfire incident:

1. *Destroyed*: Individuals whose primary residence was inside the wildfire perimeter and was destroyed.

2. *Survived*: Individuals whose primary residence was inside the wildfire perimeter and was not destroyed.

3. *Adjacent (0–10 km)*: Individuals whose primary residence in the year of the fire was between 0 km and 10 km *outside* of the wildfire perimeter. To construct this group, we draw from individuals who are included in a 10% random sample of the pooled American Community Surveys (ACS) from 2005–2022.[5]

4. *Adjacent (10–25 km)*: Individuals whose primary residence in the year of the fire was between 10 and 25 km outside of the wildfire perimeter, drawn from the ACS similar to the closer adjacent group.

5. *Other Commuting Zones*. A random sample of individuals whose primary residence in the year of the fire was in an unaffected commuting zone in the same state as the fire.

Our focal comparison group for most of the analysis is individuals in the Adjacent (10–25 km) group. In some analyses we use the Adjacent (0–10 km) or Other Commuting Zones individuals as alternative comparison groups to examine spillovers or the sensitivity of the results.

As an example of the resulting data for a single wildfire, Fig. 1 shows destroyed, surviving, and adjacent homes for the 2020 Glass Fire near Santa Rosa, California.[6] The colored markers indicate the locations of destroyed, surviving, and adjacent homes in and around the wildfire perimeter, while the dotted and dashed lines mark the 10 and 25 km thresholds used to define adjacent comparison groups. The burned area abuts the northeastern edge of Santa Rosa, the shaded urban area to the southwest. This geographic pattern of burned areas on the peripheries of urban areas is characteristic of major wildfire events.

The final analysis dataset includes approximately 1.4 million individuals living in 800,000 homes for around 150 wildfire events. Of these, approximately 32,000 were occupants of 16,000 destroyed homes and 69,500 were occupants of 30,500 homes that were exposed but survived. Adjacent (0–10 km) includes 147,000 individuals across 106,000 homes, Adjacent (10–25 km) group includes 552,000 individuals across 402,000 homes, and Other Commuting Zones is 595,000 individuals across 247,000 homes.[7]

5. Each annual ACS is a random 1% sample from the full population of housing units, making the sample we use here approximately a 2% random sample from the full population of households in these areas.

6. For plotting representative adjacent home locations, we draw a 2% random sample of single family homes from Napa and Sonoma counties' public parcel data. The resulting sample in the plot is for illustration and does not exactly match the analysis sample for confidentiality reasons.

7. Here and throughout the paper, we report aggregate statistics which are rounded following Census disclosure requirements.

3 The Distribution of Wildfire Hazard and Vulnerability

We begin by using the data to establish new descriptive facts about individuals' exposure to catastrophic wildfires. Section 3.1 compares occupants of homes within wildfire perimeters to the overall US population and to nearby unaffected homes in order to understand social and economic correlates of wildfire hazard. Sections 3.2 and 3.3 disaggregate exposed individuals into occupants of destroyed vs. surviving homes in order to understand correlates of wildfire vulnerability (that is, structure resilience).

3.1 Who Experiences Wildfires?

Table 1 reports on how wildfire-affected populations compare to the various reference populations using Census data. The first two columns summarize nationwide means for all individuals and for those living in single family homes.[8] The next three describe the comparison groups used in the empirical analyses: individuals living various distances outside wildfire perimeters and the "Other Commuting Zones" sample of individuals in unaffected commuting zones in the same state as the fire. The last three columns describe the wildfire-affected populations: Column (6) includes all occupants of homes located within a wildfire perimeter, Column (7) is just the subset of occupants whose homes were destroyed, and Column (8) includes the remaining occupants whose homes survived. All reported statistics come from the year before the fire that individuals were exposed or adjacent to.

Comparing people living in burned areas at the time of the wildfire to the U.S. population and to nearby unburned areas, we find that occupants of affected homes have an average pre-fire adjusted gross income of $192,000, notably higher than the nationwide average of $106,000 for all individuals and $119,000 for occupants of single family homes. Adjusted gross incomes for those experiencing wildfires are also higher than just outside the burn area, where average AGI is $139,000 and $135,000 for areas 0–10 and 10–25 km away, respectively. This fine-scale geographic variation is consistent with the geographic pattern illustrated for one example fire in Fig. 1, where the burned area covers an affluent area outside of the urban core. The same patterns exist when considering only wage earnings, although the difference between affected and adjacent is smaller.[9]

The racial and ethnic composition of wildfire-affected households also differs from the U.S. population and from adjacent unaffected areas. The white population share among

8. Overall national and single family homes estimates are derived from a dataset constructed by linking the American Community Survey to IRS 1040 and W-2 data similar to our main sample.

9. Differences between AGI and wages in Table 1 reflect the fact that AGI contains non-wage income sources and is a household-level measure associated with a single "primary" filer, while W-2 wages are reported at the individual level for all household members.

wildfire-affected individuals is 77%, compared to 68% for U.S. single family homes and 53% 0–10 km outside the burn perimeter. Individuals in affected areas tend to be older: the average age is 48, compared to 40 in the general population, 41 among single family home occupants, and 46 in both the 0–10 km and 10–25 km adjacent groups. Most exposed and adjacent individuals own their primary home, and about 12% of individuals across groups changed employers in the year prior to the wildfire.

In summary, individuals exposed to wildfires have been wealthier, more likely to be white, and slightly older on average than the U.S. population and individuals living just outside burned areas (though the differences narrow considerably for the last group). At the same time, while average income in the wildfire-exposed population is relatively high, there is a wide range of incomes among wildfire victims, with 25% having incomes below $42,000 (as we show below in Fig. 2).

3.2 Measuring the Vulnerability Gap

The final two columns of Table 1 consider correlates of structure destruction during wildfire events. Column (7) describes individuals whose homes were destroyed and Column (8) describes those whose homes were inside burned areas but were not destroyed. Pre-fire incomes differ substantially across the two groups at $133,000 and $220,000, respectively. A similar gap exists when considering wage earnings only. We henceforth refer to this difference in pre-fire income between occupants of destroyed and surviving homes as a "vulnerability gap". Unlike income, other pre-fire characteristics differ little between occupants of destroyed and surviving homes. Average age is similar, while occupants of destroyed homes are slightly more likely to be white.

Going beyond the mean values in Table 1, Figure 2 summarizes the empirical distributions of pre-fire income for destroyed and surviving homes. The round orange markers show means within 20 bins of pre-fire AGI for occupants of destroyed homes. For about 25% of destroyed-home occupants pre-fire AGI is below $42,000, and for about 75% it exceeds $148,300. Mean pre-fire AGI for the highest-earning 5% of destroyed-home occupants was $835,400. The triangular purple markets show distributions for occupants of surviving homes. The income gap between the destroyed and surviving groups is present throughout the distribution. Median pre-fire income is $82,000 for occupants of destroyed homes vs. $115,000 for surviving, a 40% difference. The wealthiest 5% of surviving-home households have mean pre-fire AGI of $1,500,000.

3.3 Within-Fire Comparisons

The descriptive evidence of a vulnerability gap in Table 1 does not explain *why* homes occupied by lower-income households are more likely to burn during wildfires. This section explores potential drivers of this gap using regression analysis. To ensure empirically-meaningful sample size when making within-fire comparisons, we limit the sample for these regressions to the 20 largest fires by number of homes affected. We also limit the sample to homes located inside the burned area, so that we focus only on differences between destroyed and surviving homes.

Table 2 reports results from three separate regressions of pre-fire AGI on an indicator for Destroyed. Column 1 uses all of the variation in the data, repeating the simple mean comparison from Table 1. The resulting mean difference of $87,190 reflects both within-fire variation in survival and across-fire correlation between destructiveness and average area income. Column 2 adds incident fixed effects to the regression. Controlling for incident effects recovers the average gap in income *within* wildfire incidents. This within-fire estimate is substantially smaller than the overall gap, implying that most of the vulnerability gap is explained by across-incident correlations in destructiveness and local incomes. However, the within-fire income gap of about $24,000 is still economically and statistically significant. Finally, Column (3) includes granular incident-by-street fixed effects and additional control variables for lot size and age of home. These fine-grained controls for local fire intensity within the burned area and individual property characteristics further shrink the estimate, but there remains a roughly $20,000 gap in income between destroyed and surviving homes that is marginally statistically significant. This residual difference represents the gap in vulnerability that cannot be explained by differentials in overall exposure (e.g., more destructive fires tending to occur in lower average-income areas), neighborhood exposure (e.g., wealthier neighborhoods within a given fire perimeter being located farther from high-risk areas), or observable property characteristics that may differ across households in ways that correlate with income.

4 The Effects of Wildfire Property Loss on Households

We now turn from descriptive analysis of victims' pre-fire characteristics to the effects of wildfire exposure on post-fire economic outcomes. Section 4.1 grounds the analysis in a simple model of household economics. Section 4.2 documents the empirical approach we use for estimation and Section 4.4 reports the results.

4.1 The Household Economics of Disaster Loss

This section develops a framework to put our empirical analysis in the larger economic context. Consider a household that chooses consumption c_t and labor l_t in two periods. The household maximizes utility from consumption and leisure $1 - l_t$.

$$\max_{c_1,c_2,l_1,l_2} u_1(c_1, 1 - l_1, D\alpha) + u_2(c_2, 1 - l_2) \tag{1}$$

$D \in \{0, 1\}$ is a binary random variable representing whether or not a disaster occurs at the start of period 1. α represents non-monetary disutility from the disaster event (e.g., psychological distress). Wealth includes exogenously-given initial assets m, monetary damages $D \times L$, insurance payouts $D \times I$, and per-period labor income $y_1(l_1; D)$ and $y_2(l_2; D)$. The dependence of income on the disaster captures the possibility that disasters may affect wages, ability to work, and/or commuting costs. The household borrows or saves at interest rate r. The resulting budget constraint is,

$$c_1 + \frac{1}{1+r}c_2 = m - D(I - L) + y_1(l_1; D) + \frac{1}{1+r}y_2(l_2; D) \tag{2}$$

Even in this stylized model, empirical assessment of the utility effects of a disaster is challenging. Consider, for example, the equivalent variation (EV), or the income reduction in the no-disaster state that would yield the level of utility in the disaster state. Calculating the EV requires knowledge of the household labor supply curve, the shape of consumption utility, the asset endowment, non-monetary disaster disutility, and other parameters. Such a calculation is beyond the grasp of this study and for the most part beyond the disaster literature more broadly.

We focus instead on a more modest object: the effect of the disaster on the present value of lifetime income and assets. To simplify notation, let y'_t and y^0_t represent realized income in period t with and without the disaster. The difference in the present value of lifetime income between the disaster and no-disaster states is

$$\underbrace{-L + I}_{\substack{\text{property damages} \\ \text{net of insurance}}} + \underbrace{y'_1 - y^o_1}_{\substack{\text{short-run} \\ \text{income change}}} + \frac{1}{1+r}\underbrace{(y'_2 - y^o_2)}_{\substack{\text{long-run} \\ \text{income change}}} \tag{3}$$

The disaster affects the budget through three channels. First, it reduces assets by the cost of property damages net of insurance. Second, it may change period 1 income by changing the returns to work and/or labor supply. Third, the disaster may also change period 2 income if it has long-run effects on returns to work or labor supply.

Our data allow us to measure income changes up to four years after wildfire events, illuminating the short-run effect $y_1' - y_1^0$ and, to some extent, the longer-run effect $y_2' - y_2^0$. We do not observe L, the cost of property damages, or I, insurance payments. However, a repeated finding in case studies of wildfires is that $L > I$, such that the net property damages from the event leave households worse off, even before considering the psychological and other non-monetary costs of property loss.[10] Another advantage of the tax data we use is that they include withdrawals from retirement accounts and many forms of government transfers, including unemployment insurance. Thus, the measured reductions in income include these additional channels that households may use to buffer the financial impacts of property destruction.

4.2 Empirical strategy to identify the effect of wildfire property loss

Credibly establishing the effect of a wildfire on household finances requires identifying a valid counterfactual for what would have happened to finances in the absence of the incident. Our approach is to track the evolution of outcomes for wildfire-affected individuals relative to various comparison groups in the years before and after the wildfire event. The comparison group in our main specifications is individuals living just outside the wildfire perimeter at the time of the fire. We "stack" the affected and adjacent households for each fire and measure how the average difference between affected and adjacent households evolves over time. Each affected individual experiences only one wildfire event.

Immediately adjacent homes are an attractive comparison group because this design can flexibly absorb arbitrary trends in local labor markets, sharpening identification. At the same time, it is theoretically possible that wildfires may introduce area-wide changes in wages or other labor market dynamics, which would not be captured by the highly local design in our preferred specification. We test for this directly with auxiliary analyses that compare affected households to the "Other Commuting Zones" group of homes in labor markets far from the burned area.

A key goal of our paper is to separately measure effects on occupants of destroyed and surviving homes. This facet of the design deserves discussion given the evidence in the preceding section and in other work (see, e.g., Syphard et al. 2012; Plantinga, Walsh, and Wibbenmeyer 2022; Baylis and Boomhower 2026) that property destruction is not fully random within wildfire perimeters. Our empirical design neutralizes differences in lev-

10. Cookson, Gallagher, and Mulder (2025) use policy-level data to show that 74% of destroyed homes in the devastating 2021 Marshall fire in Colorado had insurance policy limits below reconstruction cost. Klein (2018, 2023) report more broadly on the prevalence of and reasons for low coverage limits among wildfire victims. Sastry et al. (2024) use policy-level data to show that the large majority of U.S. homeowners have coverage amounts below replacement cost.

els between the various treatment groups. The primary remaining threat to identification would be differential trends in outcomes between occupants of destroyed homes, surviving homes, and the comparison groups. We address this threat by scrutinizing pre-fire trends in the various groups, finding little evidence of systematic differences.

The last remaining threat to identification would be an unrelated sudden shock to economic outcomes that occurs in exactly the same year as the wildfire. The unpredictable timing and location of catastrophic wildfires and the fact that we study 100+ wildfires across many years and states makes this kind of residual confounding highly unlikely.

4.3 Estimation

We estimate the response of various economic outcomes to wildfire occurrence using a panel local projections model following Jordà (2005) and Jorda and Taylor (2025). As described by Dube et al. (2023), an advantage of the local projections model is that it naturally accommodates recent insights about two-way fixed effects estimators with varied treatment timing (e.g., Callaway and Sant'Anna 2021; De Chaisemartin and d'Haultfoeuille 2020). The model considers how a fire in year t affects outcomes in year $t + h$, where h is between -4 and 4. Our preferred specification includes controls X_i as well as fire f-by-year t fixed effects and takes the form:

$$y_{i,t+h} = \beta_h^D 1[\text{Destroyed}]_i + \beta_h^S 1[\text{Survived}]_i + X_i + \phi_{ft} + \varepsilon_{i,t+h} \tag{4}$$

We run these as separate regressions for each value of h (excluding $h = 0$ for migration outcomes and $h = -1$ for income outcomes). The estimates we report are the β^D and β^S for each value of h. The β_h capture changes in the outcomes relative to the chosen comparison group for each fire (Adjacent homes 10–25 km away in the main specification). We include fire-by-calendar year fixed effects as a baseline, but we test for the sensitivity of the estimates to other reasonable choices of fixed effects. We report clustered standard errors by individual-event.

As mentioned above, this design leverages the spatial and temporal unpredictability of wildfire occurrence. In the event-by-calendar year fixed effects specification, the fixed effects absorb arbitrary trends in economic outcomes at the location of each fire, with the β_h coefficients capturing differential outcomes for the destroyed and survived groups relative to the comparison groups. The β_h estimates for $h < 0$ provide tests of pre-trends. The addition of direct controls X_i in some specifications helps ensure against any remaining confounding variation.

4.4 Results

4.4.1 Income

We first look at how a wildfire affects income on average across all individuals residing inside the wildfire perimeter. Fig. 3 reports the average effect on the log of Adjusted Gross Income, pooling together occupants of destroyed and surviving homes. This specification mimics what would have been possible without the individual-level damage data (while still using individual home locations and tax filing information). The estimates are near zero in all pre-fire years, implying that there were not differential trends in income among the exposed populations and the adjacent comparison group in advance of the wildfire. In the year of the wildfire (year 0) there is a small decrease in income. Most wildfires in the dataset occurred in late summer or early fall, and thus affected only a portion of year 0 earnings. In years 1 and 2, the exposed group sees a roughly four percentage point decrease in income on average, and modest effects persist through the third post-fire year before returning to zero in year four.

Pooling occupants of destroyed and surviving homes masks important differences across these groups, as shown in Fig. 4. Panel A shows that the average effects on income within the fire perimeter are almost entirely explained by large income reductions among individuals who suffered a property loss. The income reductions for this group persist for three years, peaking at 11% in the second year and returning to zero in the fourth year. The occupants of surviving homes fare much better in terms of post-fire income, with little departure from pre-fire income.

Panel B shows that results are similar when we consider only wage earnings.[11] Again, wages drop around 4% in the year of the fire, and then more than 10% in the year after. Wages climb back up in subsequent years: by year 4 we cannot reject that wages are statistically the same as in the comparison group.

The effects in Figure 4 are the difference between occupants of properties in the burned area and properties immediately outside. Particularly destructive wildfires could also have economic effects that extend beyond the fire perimeter. Figure 5 compares estimates of the effect on earnings using the focal comparison group (the 10–25 km Adjacent group, in yellow) to the two alternative comparison groups: the 0–10km Adjacent group (green) and the Other Commuting Zones group (purple). The results are remarkably similar, implying limited local wage effects of the wildfire events in our data. We return to this point in Section 6.

11. We use the inverse hyperbolic sine of wages in Panel B due to zero values in the wage data.

Finally, Fig. 7 reports separate effects on income for occupants of destroyed homes who were homeowners vs. renters at the time of the wildfire. Post-fire income losses are primarily concentrated among homeowners, consistent with disruptions associated with managing the rebuilding of a burned house. Pre-fire homeowners still have depressed earnings four years after the fire, while earnings decreases for pre-fire renters are more limited and vanish by year two or three.

4.4.2 Employment and Assets

Fig. 6 broadens the analysis to other financial and employment outcomes. There is little measurable impact on the probability of any employment (panel A) and on the likelihood of job change (panel B). These limited employment effects suggest that the observed income losses may result from a reduction in hours worked as opposed to job separations.

Occupants of destroyed homes also draw down tax-deferred retirement savings (panel C), as evidenced by data from IRS Form 1099-R. Notably, typical penalties for these withdrawals are waived by law in the wake of disaster events.[12] We note that the income estimates in Fig. 4 include these withdrawals (i.e., they are counted in AGI), which indicates that on net retirement withdrawals fall short of making individuals whole with respect to their counterfactual no-fire income.

Panel D shows immediate impacts on housing tenure for occupants of destroyed homes, as occupants of destroyed properties shift to living in rental housing before eventually returning to owner-occupied residences.

4.4.3 Migration

We observe a measure of household location every year in the EIF data. In most cases, this is based on the address used for tax filing as described in Voorheis et al. (2026). Fig. 8 reports effects on long-distance migration, which we define as moves to another county (panel A) or state (panel B). The outcome is an indicator variable equal to one if the individual does not live in the same state/county as in the year before the wildfire. Thus, it is equal to 0 for all individuals in period -1 by construction. In years prior to -1, the figure shows that individuals in the Destroyed and Survived groups are both differentially likely to not yet live in the wildfire location, relative to the Adjacent comparison group. The difference in pre-period *trends* in this event study is a mechanical result of *level* differences in the

12. Waivers of early withdrawal penalties apply throughout the county where the disaster takes place, also changing withdrawal incentives for the Adjacent groups. For that reason, panel C shows results using the Other Commuting Zones comparison group instead.

probability of moving each year; it does not imply differential trends in annual move rates in the pre-period.[13]

During the four years prior to the wildfire, the Survived and Destroyed groups are both 4–5 percentage points more likely than the Adjacent (10–25 km) group to have lived in a county other than the one where the event occurred. After the wildfire, this difference jumps to about 12 percentage points for the destroyed group. Four years post-fire, the share of Destroyed occupants in another county is still larger than the Adjacent group by about the same amount. Occupants of survived homes do not show the same migration response. Panel B shows analogous results for cross-state moves. Occupants of destroyed homes have a roughly four percentage point higher likelihood of living in a different state in the years following the wildfire.

Overall, wildfire damage appears to drive significant but not overwhelming effects on mobility for occupants of destroyed homes. About 10% of these individuals move counties as a result of the incident, and about 4% move to another state. For those that do move, these effects appear persistent across post-fire years.

5 Neighborhood Changes

Our final research question asks how the demographic composition of the area affected by a wildfire changes in the years following the event. To answer this question, we shift from following exposed individuals before and after the fire to examining addresses within the burned area perimeter. We look at how long it takes for addresses to become re-occupied after a wildfire (Section 5.1) and how average occupant characteristics evolve among homes that are occupied in the years following the fire (Section 5.2).

5.1 Return to Occupancy

Fig. 9 shows that in pre-wildfire years, more than 90 percent of addresses across all categories – destroyed, survived, adjacent, and Other Commuting Zones – were claimed as a primary address on a tax filing in the years prior to the fire. We take this tax filing behavior as a proxy for structure occupancy – however it should be noted that activity that creates an administrative record, like filing a tax return, may only imply that a mailing address still exists, and not that the structure is necessarily occupied. Following the wildfire, all groups except for the Destroyed homes continue to show similar occupancy rates. Destroyed ad-

13. The pre-trend in the event study occurs because the plotted outcome is "Not in year −1 location", which is zero by construction in period −1. Level differences in mobility lead the various groups to drift from 0 at different rates in other periods.

dresses see a roughly 50% decrease in our proxy for occupancy that persists for four years. In year five, there is some evidence of greater return to occupancy. These results suggest long delays in rebuilding and re-occupying wildfire- damaged homes. At a neighborhood level, this implies large persistent shares of vacant, damaged properties.

5.2 Occupant Characteristics Among Occupied Homes

Finally, we examine post-wildfire changes in the demographics among occupied homes. Any compositional changes observed in the post-wildfire period reflect a combination of differential speed of rebuilding among pre-wildfire residents with divergent characteristics and/or changes due to permanent in- and out-migration.

Fig. 10 shows that the composition of individuals living in fire-destroyed addresses changes notably after the fire. Panel A shows that, among destroyed homes that have returned to occupancy, mean occupant income is higher in the post-wildfire years. There is also a small increase in income among occupants of destroyed homes. Just outside the wildfire perimeter, there are small decreases in income, consistent with a small amount of income-based sorting into the burned area. Finally, there is a visible increase in income for occupants of survived homes several years after the fire. These income dynamics are consistent with changes in the post-fire housing stock reported in Issler et al. (2024).

Panel B of Fig. 10 shows the share of the population over age 65. The over-65 share decreases notably among destroyed homes, while remaining unchanged among survived homes. There is also a small increase in the over-65 share in the areas adjacent to the wildfire perimeter, again consistent with sorting of older individuals out of the burned area.

6 Discussion

6.1 Wildfire Exposure and the Vulnerability Gap

The results in Section 3 offer a new perspective on the demographics of wildfire risk in the United States. The data show that the population most exposed to this increasingly-severe peril differs from the groups most exposed to other weather disasters. Table 3 summarizes mean pre-event incomes reported in studies of other types of natural disasters, adjusted to year 2022 dollars. One interesting benchmark is the national sample of Federal Disaster Loan applicants in Collier and Ellis (2024). That dataset includes victims of many different types of disasters, though the bulk are severe storm and hurricane events. The average pre-event income among that group is $84,275, compared to $145,679 for wildfire victims.

To some extent, these differences reflect differences in nominal wages across regions of the

country. Hurricane and flood risk are concentrated in the Southeast, while many damaging wildfires occur in California (which is at the intersection of population density and physical factors that create wildfire risk). It is also important to note that while average incomes of fire-exposed populations are relatively high, there is a wide range within that group as shown in Fig. 2. Twenty-five percent of wildfire victims have incomes below $42,000.

The address-level damage data allow us to differentiate outcomes for occupants of destroyed vs. surviving homes. The clear income difference here is suggestive of a "structure vulnerability gap". Much of this gap is explained by across-incident correlations between income and property loss (more destructive fires tending to occur in lower-income locations). However, one-third of this income difference persists even when looking within individual wildfire incidents. The possibility of such a gap has been hinted at in earlier work that shows differential wildfire damage outcomes by building vintage and other factors (Baylis and Boomhower 2026; Syphard et al. 2012). The results here reinforce the value of working with individual-level damage data and underscore the importance of identifying effective vulnerability-reducing interventions.

6.2 Income Reductions for Wildfire Victims

The data show significant, persistent reductions in income for individuals who lose their home in a wildfire. For a homeowner with the mean AGI in our destroyed homes sample, the income reductions across years in Panel A of Fig. 4 imply total earnings losses of about $35,000. These earnings losses add meaningfully to the total financial hit taken by wildfire victims. Returning to the economic framework in Equation 3, we can roughly calculate how these earnings losses fit into the overall financial impact of the wildfire. Unlike some other natural catastrophe perils, wildfire is covered by standard homeowners insurance policies. To compare earnings losses to net-of-insurance rebuilding costs, we draw estimates of insurance coverage from Cookson, Gallagher, and Mulder (2025), which uses exceptional contract-level data from a large Colorado wildfire to show that homeowners insurance was adequate to cover 88% of rebuilding costs for the average fire victim. Applying that ratio to the average reconstruction cost for a California home in wildfire hazard areas of $600,000 from Boomhower et al. (2024) yields a back-of-the-envelope estimate of typical uninsured property damages of $72,000. The income reductions that we measure in this study add roughly 50% to that loss number for the typical wildfire victim.

One factor driving the persistent income losses that we measure may be that wildfire-damaged homes are typically completely destroyed and require ground-up reconstruction. Fire victims also lose essentially all of their belongings inside the home. Floods and hurricanes impose a wider range of damages, including some smaller losses with accompany-

ing shorter repair times. The median total loss amount (insured and uninsured damages) among FDL applicants in Collier and Ellis (2024) is $51,000, which implies a more limited extent of damage. The time costs of managing the reconstruction of a completely destroyed home may lead individuals to reduce labor supply at least temporarily. The disproportionate earnings decreases for pre-fire homeowners vs. renters lend some suggestive support to this channel. Individuals who owned their home prior to the wildfire experience cumulative earnings decreases equaling 39% of pre-fire AGI, while cumulative earnings losses are much smaller for occupants of destroyed homes who were renters.

6.3 Local Labor Market Impacts

The main analyses focus on narrow comparisons in- and outside of the wildfire burned area. This specification follows the real estate literature on mortgage and home price outcomes after wildfires (McCoy and Walsh 2018; Issler et al. 2024; Biswas, Hossain, and Zink 2023; Kahn, Ouazad, and Yönder 2024) as well as papers in the literature on financial impacts of other disasters (Billings, Gallagher, and Ricketts 2022; Gallagher and Hartley 2017; Gallagher, Hartley, and Rohlin 2023). A separate literature on the financial consequences of mega-disasters—in particular, Hurricane Katrina—emphasizes comparisons between affected individuals and individuals in "control" cities or counties located far from the disaster area. These comparisons are motivated by the possibility that large disasters may affect local wages (for example, by increasing demand for construction labor) and that such effects are part of the overall effect of interest. In particular, Deryugina, Kawano, and Levitt (2018) and Groen, Kutzbach, and Polivka (2020) find that Hurricane Katrina *increased* earnings among Hurricane Katrina victims several years after the storm due to increased labor demand and decreased labor supply in the affected area.

The "Other Commuting Zones" specification in Fig. 5 investigates whether such effects are present in our setting. There is no evidence of important local labor market effects from wildfires along the lines of those that followed Hurricane Katrina. Katrina was the costliest natural disaster in U.S. history, causing over $200 billion of property damage over a wide geographic area, and was followed the same year by another $40+ billion of damage from Hurricanes Rita and Wilma in nearby areas. These losses are an order of magnitude larger than the costliest disaster in our incomes analysis, the 2018 Camp Fire. The Camp Fire caused $18 billion of property damage.[14] Most wildfire incidents are substantially smaller. This difference from the Hurricane Katrina literature underscores the value of studying a large number of disasters in addition to individual exceptional loss events.

14. Damages from the 2025 Palisades and Eaton Fires in Los Angeles are currently estimated be about $40 billion, though the long-run economic impacts will not be known for several years.

6.4 The 2025 Los Angeles Wildfires

In January 2025, the Palisades and Eaton Fires in Los Angeles caused record-high amounts of property damage. These fires are not included in our main analysis dataset because their recency precludes study of post-fire outcomes. Here we examine pre-fire demographics for these events to see if the patterns from earlier fires continue to apply.

Table 4 presents an auxiliary analysis of pre-fire demographics for these events. We see that the same broad patterns from earlier wildfires hold. The Eaton Fire near Pasadena affected individuals with a moderately higher AGI than the Los Angeles County average. The population affected by the Palisades Fire near Pacific Palisades was highly affluent on average. Over 70% of individuals exposed to the Palisades Fire were white, compared to around 30% for the Eaton Fire and 23% for the rest of Los Angeles County. Fire-exposed individuals were more likely to be homeowners and about as likely to be over 65 as other Los Angeles residents. In both cases, we continue to see notable gaps in income between occupants of destroyed and surviving homes.

In summary, demographic patterns of exposure and income gaps in structure survival in the 2025 L.A. fires were similar to what we observe in other wildfires over the previous two decades. In future years our analyses can be updated to understand whether other key empirical patterns—post-fire income losses, persistent relocation effects, and neighborhood changes—also follow the patterns that we document for earlier fires.

7 Conclusion

The economic consequences of wildfire are unevenly distributed, extend beyond the direct loss of property, and continue to unfold as individuals and communities recover. We establish these facts by linking address-level damage assessments for approximately 150 U.S. wildfires to individual administrative tax and Census records. By identifying individual-level property damages and following both residents and addresses after each event, our analysis provides a more complete account of disaster incidence and recovery.

Homes affected by wildfires largely reflect the overall geography of wildfire risk, which is concentrated in the western United States, especially California. Wildfire victims are wealthier on average than victims of other natural disasters. This difference reflects both nominal wage differences between wildfire- and hurricane-exposed areas and the location of many wildfire events in affluent neighborhoods on the periphery of urban areas. Not all wildfire victims are wealthy, however. A meaningful share have incomes below $40,000. The individual-level damage data also allow us to uncover a clear difference in pre-fire

income between occupants of surviving and destroyed homes. Future research to understand the extent to which this gap reflects unexploited resilience opportunities among less affluent households will be valuable.

Wildfires cause substantial earnings losses, which are almost entirely concentrated among individuals whose homes are destroyed. Across several years following the event, the average wildfire victim foregoes an amount totaling 26% of their single-year pre-fire adjusted gross income, or $35,000. This may be a substantial fraction of the total financial effect of the incident on the household, especially given that direct property damages are often partially covered by insurance. Unlike the largest coastal hurricanes, there is no evidence that wildfires have equilibrium impacts on wages or employment for individuals who do not suffer a direct property loss. Individuals whose homes are destroyed are somewhat more likely to make long-distance moves following the wildfire. These moves appear to be persistent for at least four years.

We also measure important neighborhood-level effects in burned communities. Rebuilding is slow; many burned properties are vacant for years. There is also a measurable change in the demographics of these neighborhoods following the fire. Previously destroyed homes are re-occupied by systematically higher-income residents. The share of individuals over 65 also falls significantly. Further tracing the effects of these changes in neighborhood composition is an important area for future research.

The occurrence of catastrophic wildfires has increased dramatically over the past decade and is projected to continue to grow due to climate change, ongoing construction of homes and other assets in high-hazard areas, and an accumulation of fuel due to past fire suppression. Adapting to this escalating hazard requires understanding the economic effects of these events and their incidence across the population. Continued research on these topics is vital for coordinating investments in resilience and supporting recovery from future fires.

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Table 1: Wildfire-exposed Individuals vs. the U.S. Population

	U.S. Averages		Comparison Populations			Wildfire-Exposed Populations		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	U.S. Population	U.S. Single Family Homes	Adjacent 0-10 km	Adjacent 10-25 km	Other Commuting Zones	All Affected	Destroyed	Survived
Income								
Pre-fire AGI ($1,000)	106	119	139	135	124	192	133	220
			(372)	(449)	(345)	(-)	(313)	(1,661)
Pre-fire Wages ($1,000)	49	53	92	94	87	110	80	124
			(186)	(327)	(316)	(-)	(148)	(675)
Race and Ethnicity								
Share White	62	68	53	46	46	77	82	75
			(50)	(50)	(50)	(-)	(39)	(43)
Share Hispanic	17	15	24	29	30	9.6	9.5	9.7
			(43)	(45)	(46)	(-)	(29)	(30)
Share Asian	5.3	4.8	14	16	14	6.9	3.5	8.4
			(35)	(36)	(34)	(-)	(18)	(28)
Share Black	12	9.2	2	2.7	3.7	1.1	0.68	1.3
			(14)	(16)	(19)	(-)	(8.2)	(11)
Other Characteristics								
Age of Filer	40	41	46	46	46	48	47	48
			(12)	(12)	(12)	(-)	(12)	(12)
Pre-fire Homeownership	-	-	0.82	0.87	0.84	0.73	0.71	0.74
			(0.38)	(0.34)	(0.37)	(-)	(0.46)	(0.44)
Pre-fire Job Switching	-	-	0.13	0.14	0.14	0.12	0.13	0.12
			(0.34)	(0.34)	(0.35)	(-)	(0.34)	(0.32)
Observations								
Occupants			147,000	552,000	595,000	101,500	32,000	69,500
Homes			106,000	402,000	247,000	46,500	16,000	30,500

Notes: Dollar amounts are in year 2022 dollars. Cell entries are means, with standard deviations in parentheses. "U.S. Single Family Homes" (2) reflects nationwide statistics for occupants of single family homes. "Adjacent 0-10 km" (3) and "Adjacent 11-25 km" (4) include occupants of homes located those distances outside the burn perimeter. "Other Commuting Zones" (5) includes a random sample of occupants of homes in commuting zones in the same state as a fire. "All Affected" (6) is the occupant-weighted average of the "Destroyed" (7) and "Surviving" (8) columns; standard deviations are therefore not defined for this column. Race and ethnicity shares are percentages and are coded as mutually exclusive categories (e.g., Share White Non-Hispanic). All observation counts are rounded for disclosure reasons.

Table 2: Within-event Vulnerability Gaps in Income

	(1)	(2)	(3)
Destroyed	-87,190***	-23,650***	-19,610*
	(13,530)	(8,212)	(10,830)
Additional Controls			
Fire FE	No	Yes	No
Street-by-Fire FE	No	No	Yes
Property Characteristics	No	No	Yes

Notes: Table reports three separate OLS regressions of pre-fire adjusted gross income on an indicator for Destroyed. All columns limit the sample to homes inside the burned area for the largest 20 incidents in the dataset. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level.

Table 3: Average Pre-Event Income for Wildfire-Affected Homes vs. Other Disasters

Study	Context	Mean Pre-Event Income ($2022)
Deryugina, Kawano, and Levitt (2018)	Hurricane Katrina	54,526
Collier and Ellis (2024)	Disaster Loan Applications	84,275
Billings, Gallagher, and Ricketts (2022)	Hurricane Harvey	89,632
This study	150 wildfires	192,340

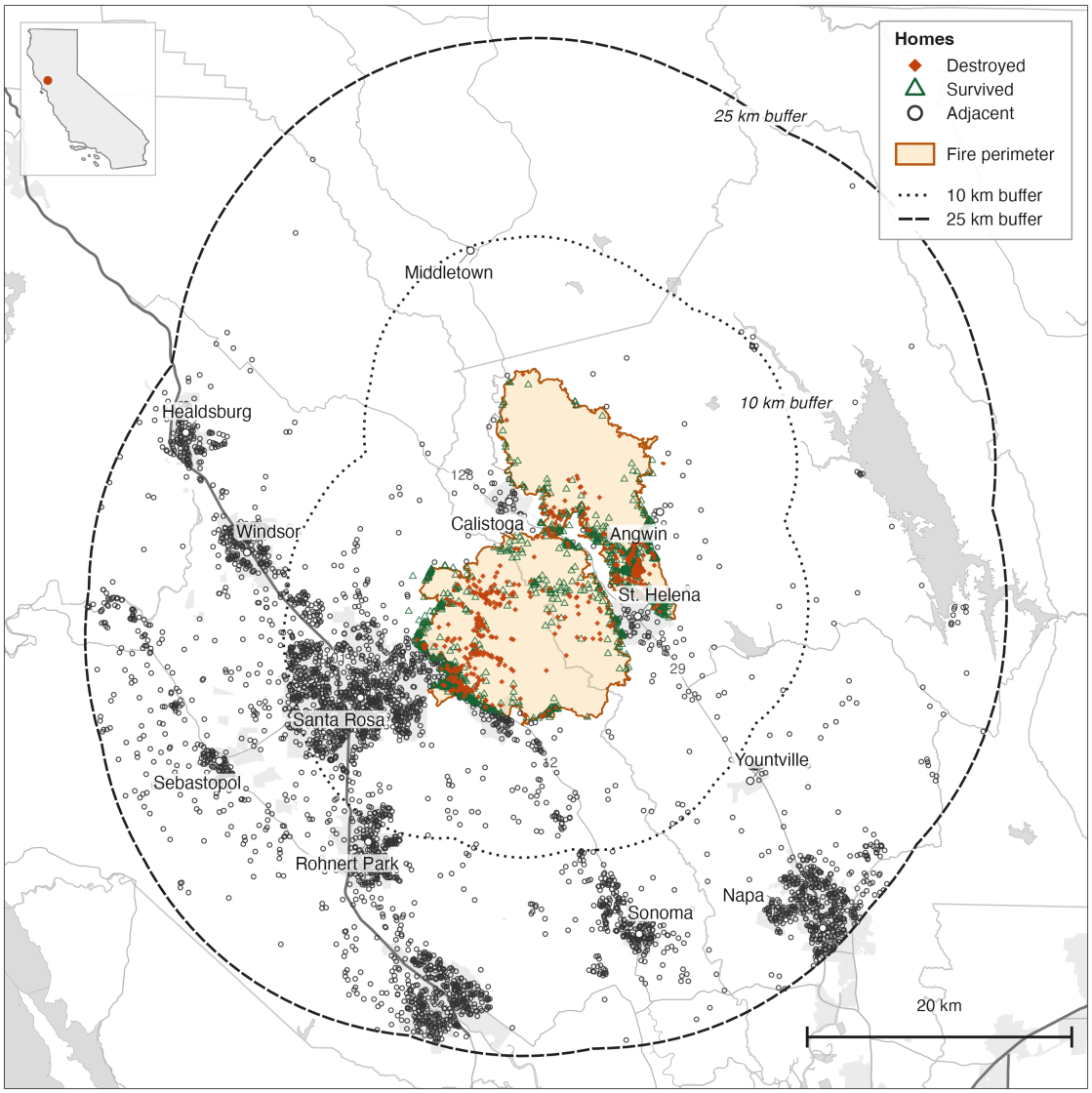
Notes: Values are reported mean incomes before the disaster event in each study adjusted to year 2022 dollars. Values in this study and Deryugina, Kawano, and Levitt (2018) are adjusted gross income.

Table 4: Exposure Demographics for 2025 Los Angeles Fires

		Eaton		Palisades	
	(1)	(2)	(3)	(4)	(5)
	Rest of LA	Destroyed	Survived	Destroyed	Survived
Income					
Pre-fire AGI ($1,000)	120	141	222	870	962
Race and Ethnicity					
Share White	23	31	38	77	71
Share Hispanic	43	30	29	5	6
Share Asian	12	5	9	6	8
Share Black	8	22	11	1	1
Other Characteristics					
Pre-fire Homeownership	0.49	0.85	0.77	0.79	0.79
Over 65	0.20	0.20	0.21	0.23	0.23

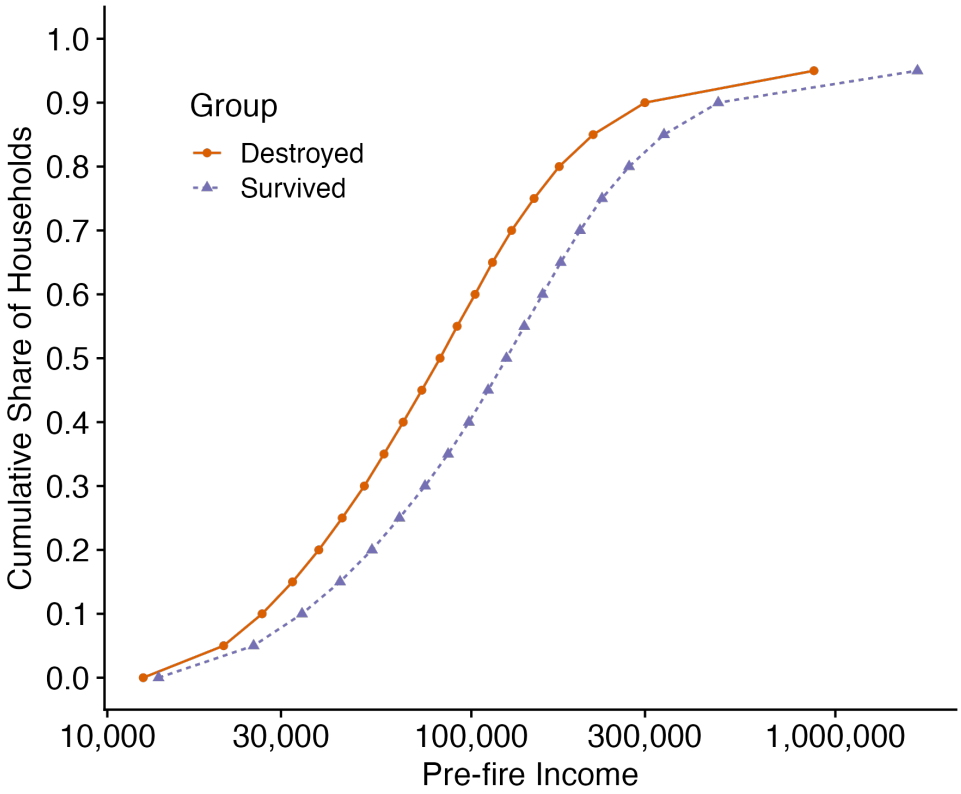
Notes: Mean pre-fire characteristics for occupants of the Eaton Fire perimeter, the Palisades Fire perimeter, and the rest of Los Angeles County. Race and ethnicity shares are percentages and are coded as mutually exclusive categories (e.g., Share White Non-Hispanic). These 2025 Los Angeles fires are not reflected in other calculations in this paper due to their recency.

Figure 1: Destroyed, Surviving, and Adjacent Homes in the 2020 Glass Fire



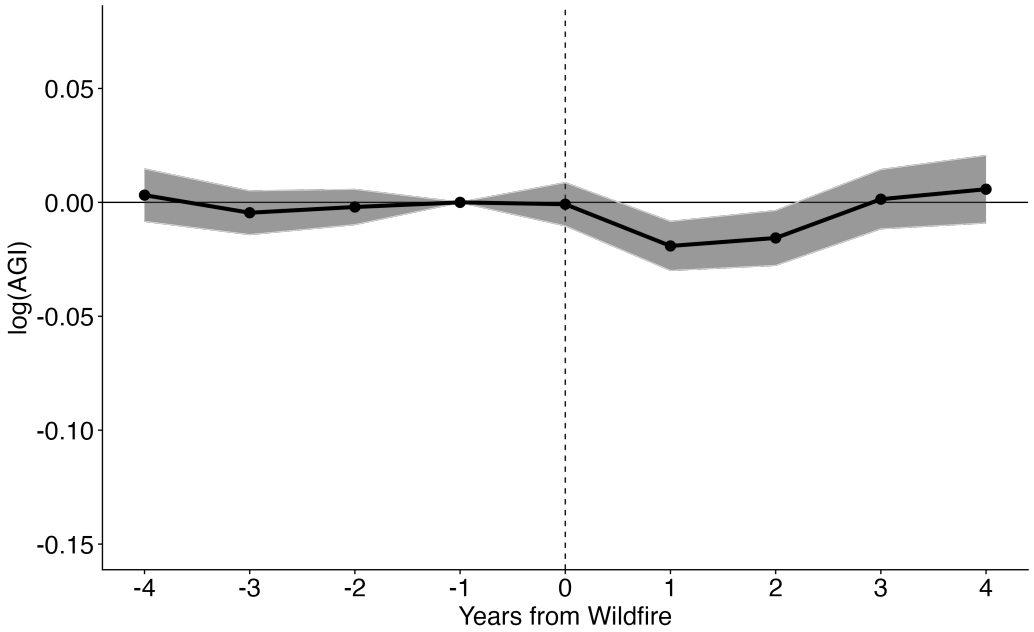
Notes: Map visualizes homes in or near the perimeter of the 2020 Glass Fire in Northern California. The wildfire perimeter is shaded in orange, and the dotted and dashed lines outside the perimeter represent 10 and 25 km adjacent areas. Colored markers indicate locations of destroyed and surviving homes as reported in public damage inspection data. Black circular markers are a 2% sample of adjacent homes from Napa and Sonoma counties' public parcel data.

Figure 2: Pre-fire Income Distributions for Occupants of Affected Homes



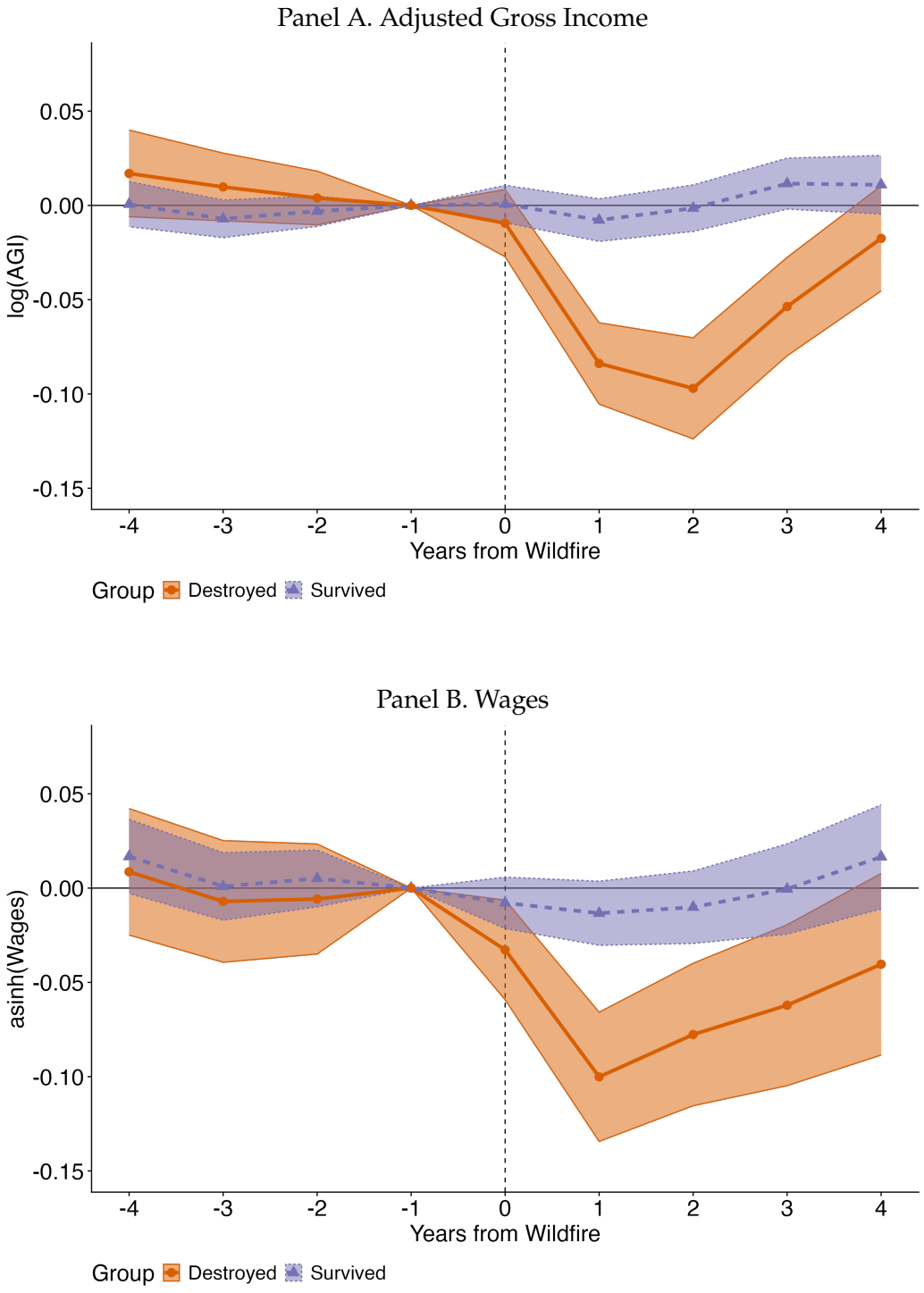
Notes: Figure reports 20 equal-observation bins of pre-fire AGI for occupants of destroyed and surviving homes (in 2022 dollars) on a logarithmic scale.

Figure 3: Effect of Wildfires on Incomes, Pooling Destroyed and Survived Homes



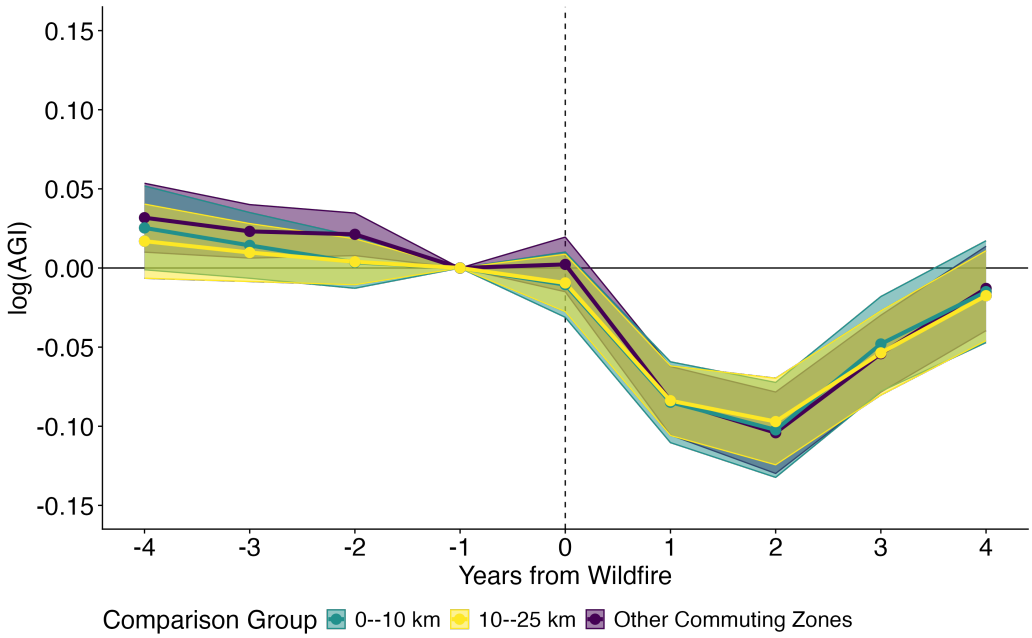
Notes: Figure reports local projection estimates of differential log adjusted gross income for occupants of homes inside the wildfire perimeter relative to the Adjacent comparison group in years before and after the wildfire.

Figure 4: Effect of Wildfires on Income by Damage



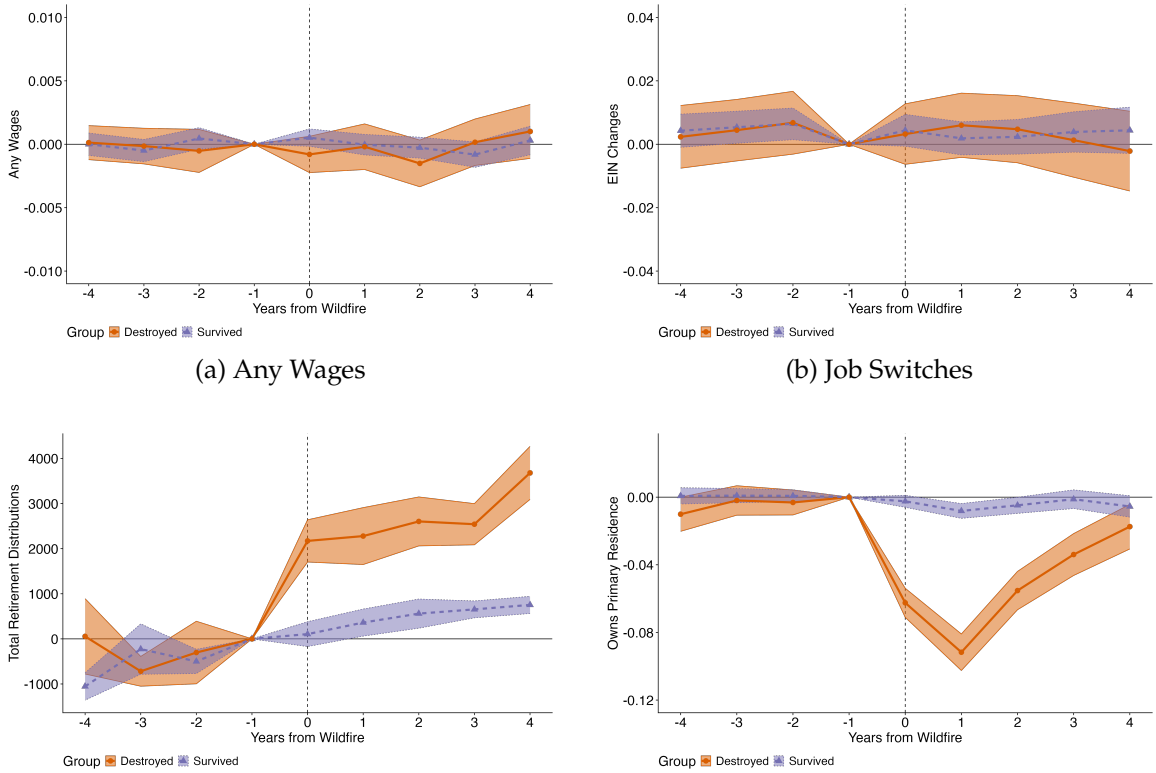
Notes: Figure documents the effect of a wildfire event on income, split by whether the individual's home was destroyed or not. Panel A reports local projection estimates of the effect on log Adjusted Gross Income for Destroyed and Survived groups relative to the Adjacent (10–25 km) group in years before and after the wildfire and Panel B reports similar local projection estimates for the inverse hyperbolic sine of wages.

Figure 5: Effect on Income Among Destroyed Home Occupants, Alternative Comparison Groups



Notes: Figure reports local projection estimates of differential log adjusted gross income for individuals in the Destroyed group relative to the Adjacent (10–25 km) group in years before and after the wildfire.

Figure 6: Effects on Employment and Asset Ownership



(a) Any Wages

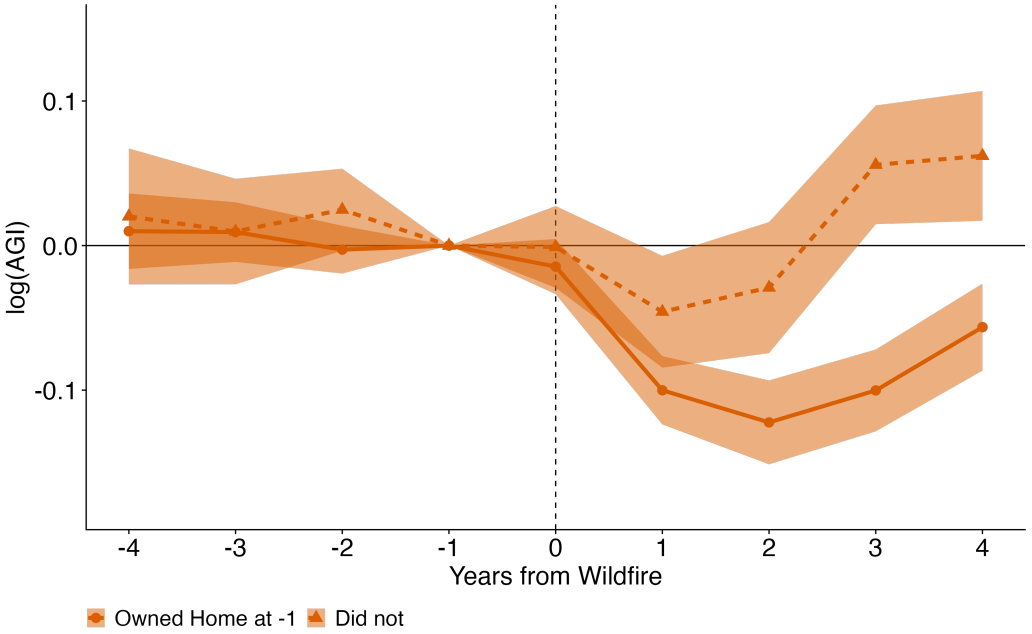
(b) Job Switches

(c) Retirement Account Withdrawals

(d) Owns Primary Residence

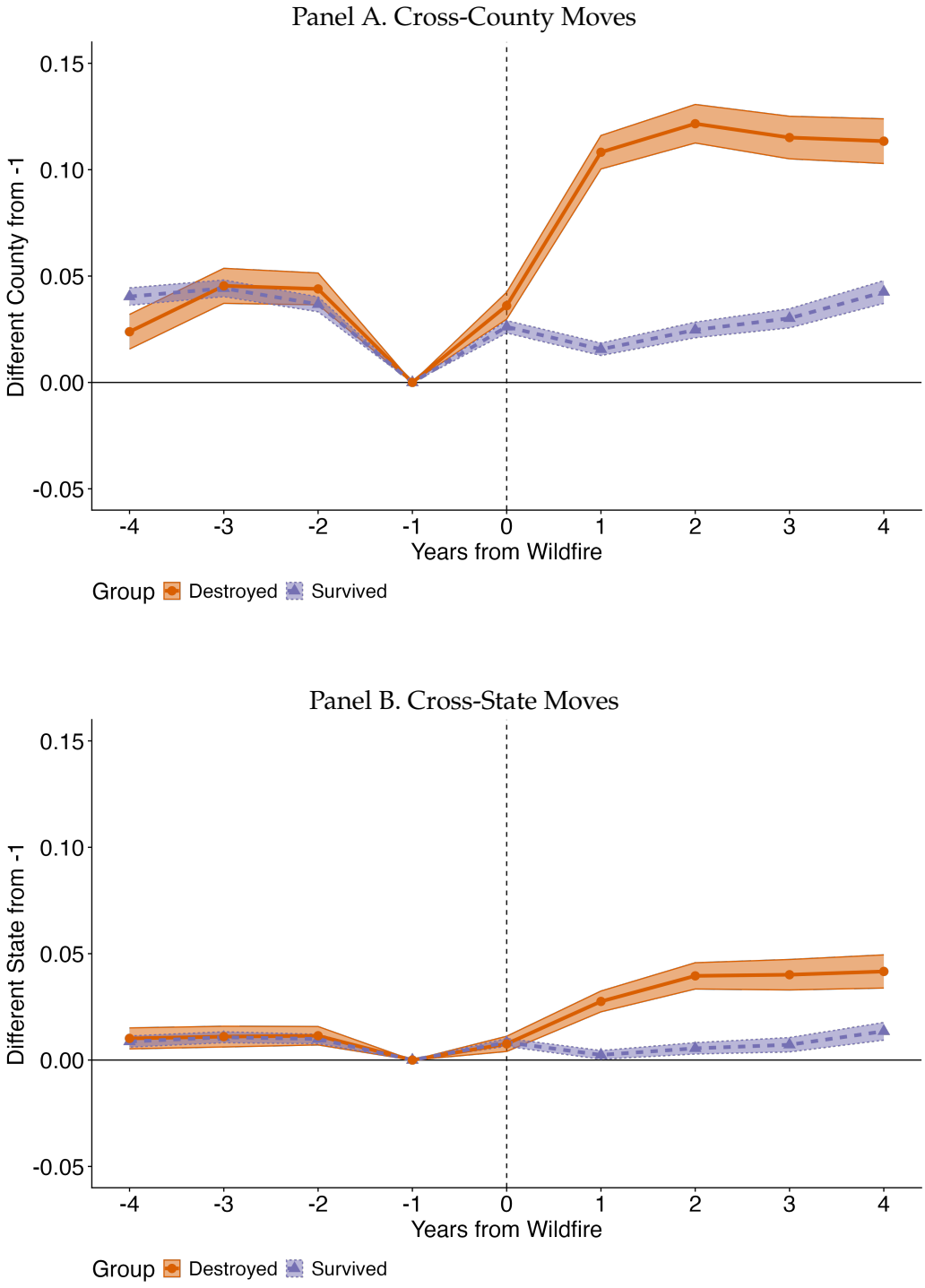
Notes: Figure shows impacts of wildfire event on various measures of employment and asset ownership. Panels (a) and (b) report local projection estimates for the probability of any wages and the probability of changing employers. Panels (c) and (d) report local projection estimates of the amount withdrawn from tax-deferred retirement accounts and share living in owner-occupied housing. The comparison group in all panels is individuals in the Adjacent (10–25 km) group, except retirement withdrawals where the comparison group is the Other Commuting Zones group. See text for details.

Figure 7: Income Effects for Occupants of Destroyed Homes, by Pre-fire Homeownership



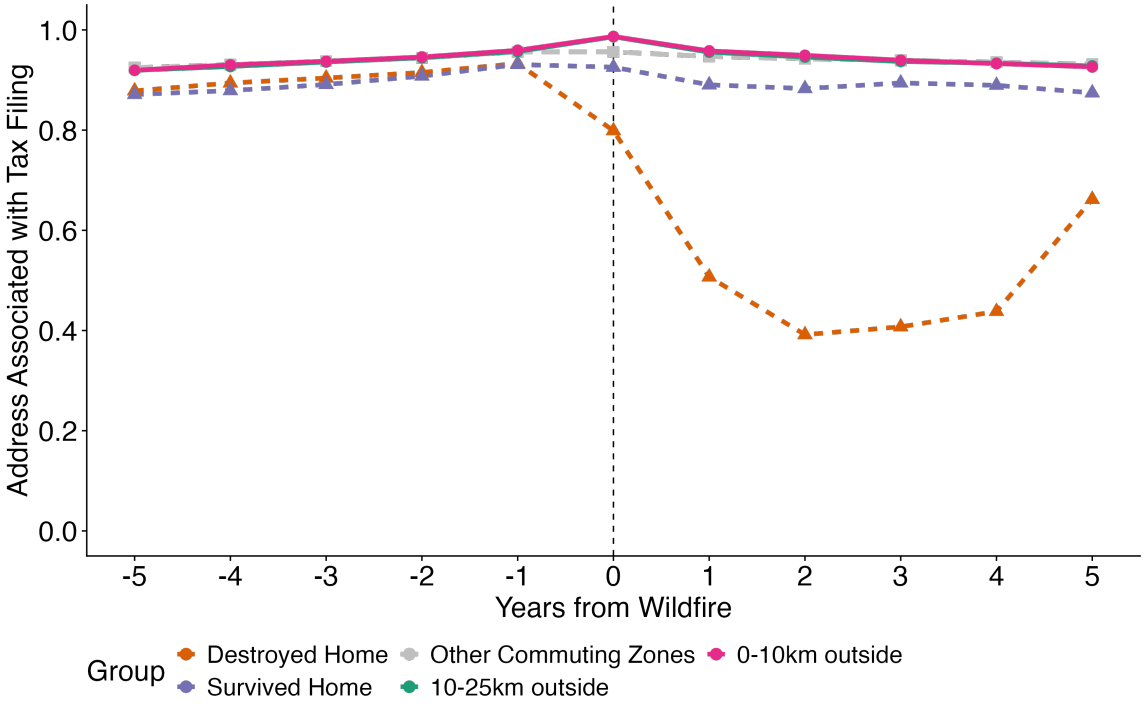
Notes: Figure shows impacts on income for occupants of destroyed homes, split by pre-fire homeownership. Each line is a set of local projection estimates of the effect of home destruction on Adjusted Gross Income (AGI) in comparison to the Adjacent (10–25 km) group, with the two sets of estimates split by whether the individual owned their home in the year before the fire. Bars display 95% confidence intervals.

Figure 8: Migration by Wildfire Victims



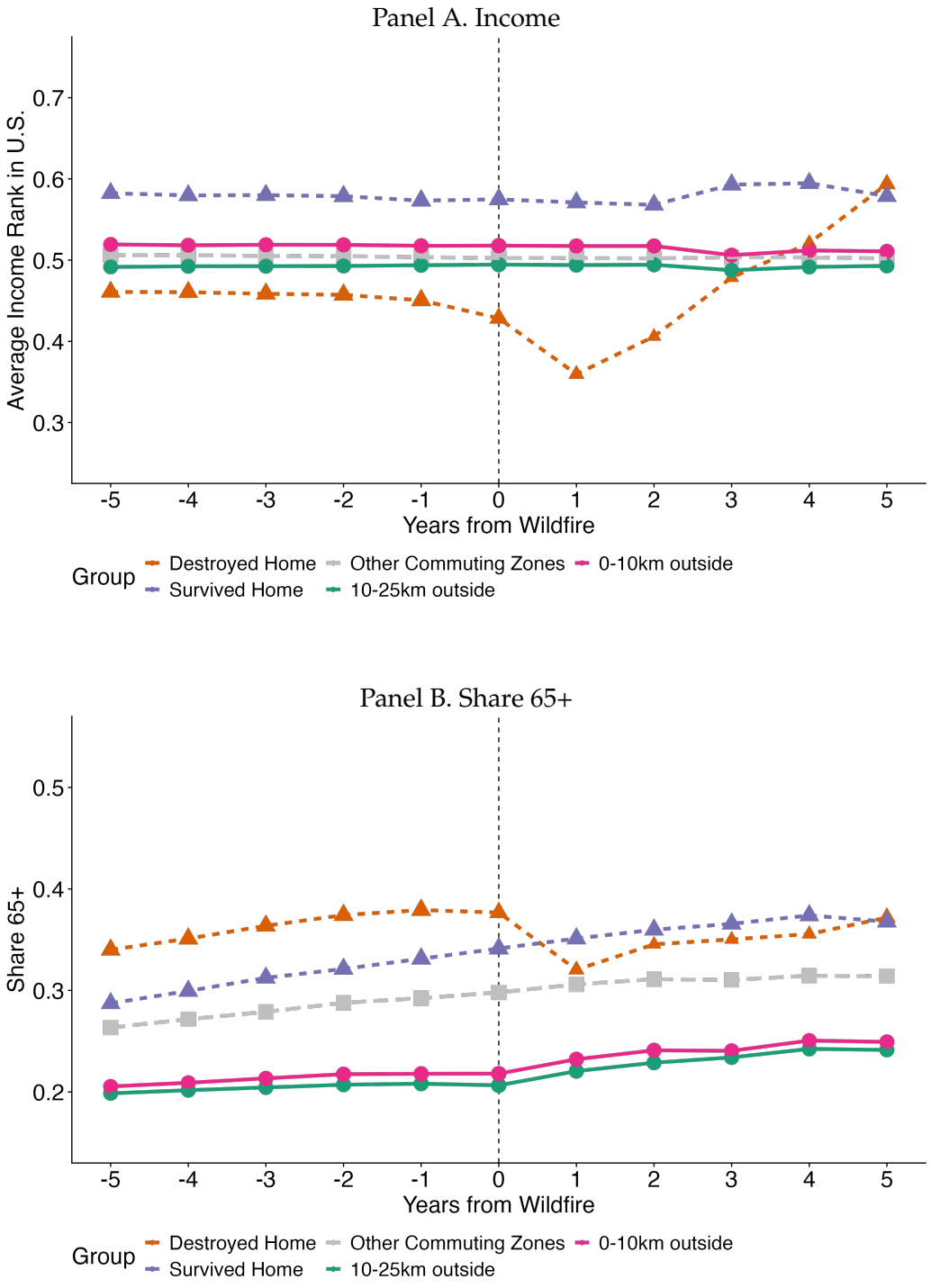
Notes: Figure reports local projection estimates of differential likelihood of living in the year −1 location for Destroyed and Survived groups relative to the Adjacent (10–25 km) group in years before and after the wildfire.

Figure 9: Share of Addresses Occupied Before and After Wildfire



Notes: Time-varying means of occupancy in years before and after the wildfire. Occupancy is defined as someone using the home as a primary address for tax filing.

Figure 10: Compositional Changes Among Occupied Addresses



Notes: Time-varying means of income and share 65+ for occupied addresses in years before and after the wildfire. Occupancy is defined as someone using the home as a primary address for tax filing. Marker sizes are proportional to the fraction of addresses that are occupied in a given year.